

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
MODEL EXAMINATION

Year/ Sem: I/I
Date: 10.02.2023

Max. Marks: 75
Max. Time: 3 Hours

PART I – (10 x 2 = 20 marks)

Answer ALL questions

SECTION – A
(ELECTRICAL)

1. Define ohms Law.
2. What is Active Element and passive Element?
3. State current division and voltage division rule?
4. What is principle of DC Generator?
5. Define Transformer and its losses

SECTION – B
(ELECTRONICS)

1. What is doping.
2. Draw the symbol of two types of JFET.
3. Draw the diagram of JK flip flop.
4. What is microwave?
5. Brief about cellular or mobile services.

PART II – (55 marks)

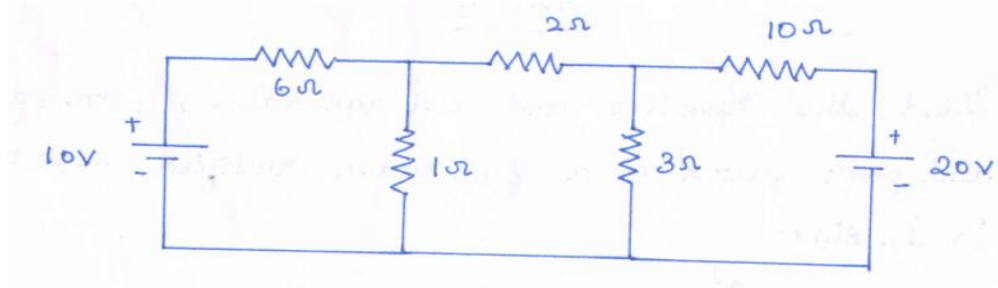
SECTION – A (ELECTRICAL) – (27 Marks)

UNIT - I

1. Deduce the expressions for star to delta transformation and vice versa.

(OR)

6. Find the current through 1Ω resistor and 3Ω in the network using mesh current analysis. (9)



UNIT – II

7. Explain the RLC series circuit. (9)

(OR)

8. Explain the two wattmeter method of measurement of three phase power with neat phasor diagram. (9)

UNIT – III

9. What are the types of 1 phase Induction motor? Explain any two types. (9)

(OR)

10. Explain with neat block diagram Hydel power plant. (9)

SECTION – B (ELECTRONICS) – (28 Marks)

UNIT - I

11. Explain the working principle of RC phase oscillator with a neat circuit diagram. (10)

OR

12. Explain the working of BJT in CE configuration with suitable diagram and characteristics curves. (10)

UNIT - II

13. Explain the working of JK flip flop. (9)

OR

14. Explain the operation of ripple counter. (9)

UNIT - III

15. How the computer networks are classified? Explain in brief. (9)

OR

16. What is earth station? Explain the satellite communication system with the relevant block diagram. (9)